AWS PROJECT

AWS Auto Scaling

with Application Load Balancer

Self-Healing and Highly Available AWS Architecture

- Auto Scaling
- Load Balancer
- High Availability
- Self-Healing

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Project Overview

🎯 Project Goal

Build a fault-tolerant, self-healing web infrastructure on AWS using Auto Scaling and an Application Load Balancer.

🔗 Extends Previous Setup

Builds on a prior ALB + EC2 project by adding Auto Scaling for dynamic capacity management and automatic recovery.

⚙️ High Availability

EC2 instances deployed across multiple Availability Zones — ap-south-1a and ap-south-1b — ensuring no single point of failure.

🔄 Self-Healing

ASG detects unhealthy instances via health checks and automatically replaces them, maintaining desired capacity at all times.

Architecture Overview

This architecture ensures zero downtime by combining load balancing with automatic instance recovery.

→ Requests arrive at the ALB DNS endpoint (public-facing)

→ ALB forwards traffic across healthy EC2 targets in the Target Group

→ ASG manages EC2 lifecycle — launches, terminates, and replaces instances

→ Health checks ensure only healthy instances serve traffic

Multi-AZ Deployment
ap-south-1a | ap-south-1b

AWS Services Used

🖥️ Amazon EC2

t2.micro instances running Apache web server. Launched automatically via Launch Template.

📈 Auto Scaling Group (ASG)

Manages EC2 fleet. Automatically replaces unhealthy instances. Min: 1 | Desired: 1 | Max: 2

⚖️ Application Load Balancer

Distributes incoming HTTP traffic across healthy EC2 targets in multiple AZs.

🎯 Target Group

Bridges ALB and EC2. Performs HTTP health checks on port 80 to filter healthy targets.

🌐 VPC & Subnets

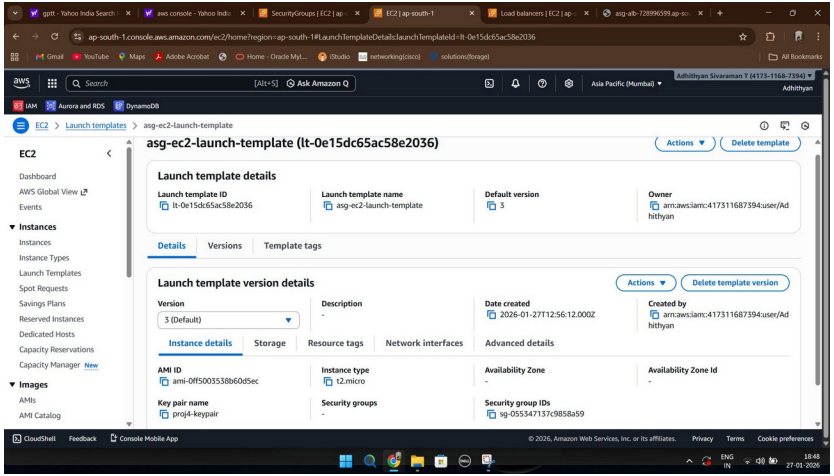
Isolated network. Public subnets in ap-south-1a and ap-south-1b for multi-AZ setup.

🔒 Security Groups

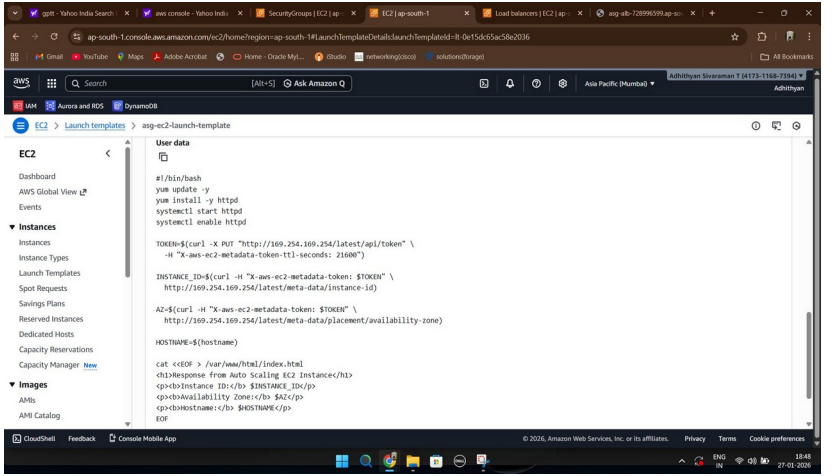
ALB SG allows port 80 from internet. EC2 SG allows port 80 only from ALB — least privilege.

Launch Template & Automation

📋 Launch Template



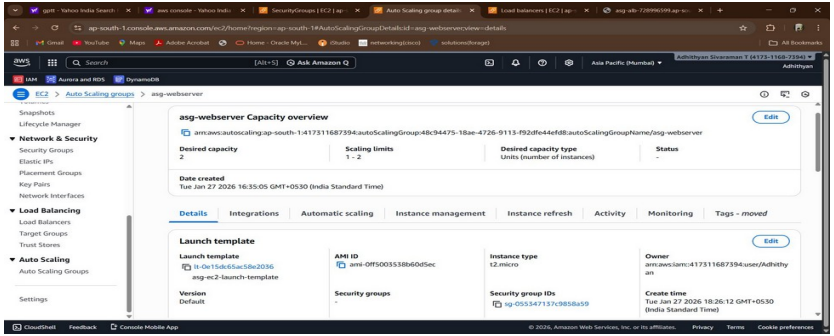
⚙️ User Data Script



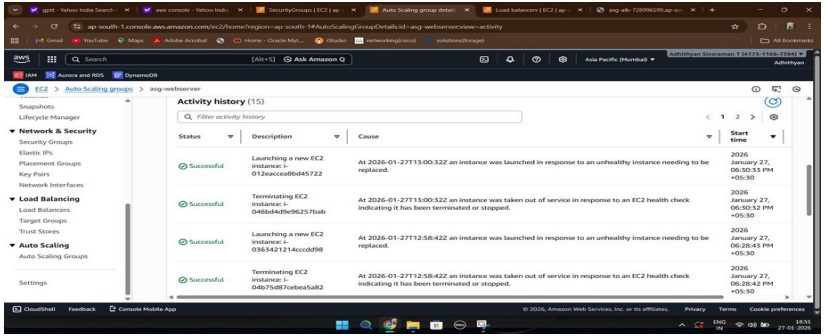
- ► AMI: ami-0ff5003538b60d5ec | Instance: t2.micro | Key: proj4-keypair
- ► User data auto-installs Apache (httpd), starts on boot, and writes a dynamic HTML page
- ► Page displays Instance ID, AZ, and Hostname — fetched from EC2 Instance Metadata Service (IMDSv2)

Auto Scaling & Self-Healing

ASG Configuration



Scaling Activity Log



Capacity Settings:

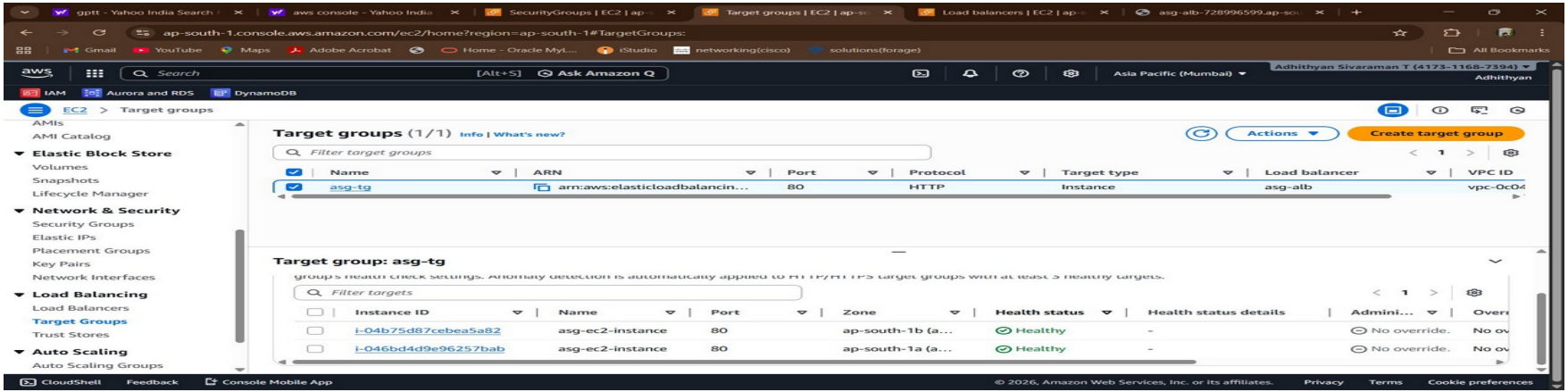
Min 1 | **Desired 1** | **Max 2**

Activity log confirms ASG replaced unhealthy instances - self-healing in action

New instances use same Launch Template - ensures configuration consistency

Load Balancer & Health Checks

🎯 Target Group — Both Instances Healthy



⚖️ Traffic Distribution

ALB routes requests across both EC2 targets using round-robin. Load is evenly spread across AZs.

❤️ Health Checks (HTTP /)

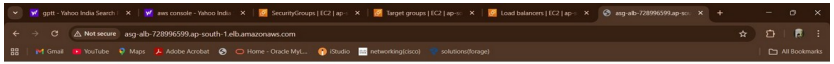
ALB sends periodic HTTP GET / requests to each target. Unhealthy targets are removed from rotation.

🛡️ Healthy Traffic Only

Only instances passing health checks receive live traffic. Unhealthy ones are quarantined automatically.

Proof of Load Balancing

Instance 1 — ap-south-1b



Response from Auto Scaling EC2 Instance

Instance ID: i-0363421214ccccdd98

Availability Zone: ap-south-1b

Hostname: ip-172-31-6-88.ap-south-1.compute.internal

Instance 2 — ap-south-1a

Response from Auto Scaling EC2 Instance

Instance ID: i-012eaceea8bd45722

Availability Zone: ap-south-1a

Hostname: ip-172-31-27-128.ap-south-1.compute.internal

Refreshing the same ALB URL returns different Instance IDs and AZs — confirming active load balancing across two Availability Zones.

Each refresh hits a different EC2 instance, proving real-time traffic distribution via ALB.

Full screenshots, configuration steps, and architecture notes are available in the GitHub repository.

Key Concepts

☆ RECRUITER HIGHLIGHT

🔄 Auto Scaling = Self-Healing

ASG continuously monitors instance health and replaces failed nodes without manual intervention.

🗑️ Disposable Instances

Instances in an ASG are ephemeral. Any failed instance is automatically terminated and a fresh one is provisioned.

🎯 Target Group as Bridge

The Target Group decouples the ALB from EC2. It translates routing rules and tracks individual instance health.

❤️ Health Checks Drive Routing

ALB only forwards traffic to instances that pass HTTP health checks — unhealthy ones are silently removed.

🔒 Least Privilege Security

EC2 Security Group allows port 80 only from the ALB Security Group — never directly from the internet.

🌍 Multi-AZ = High Availability

Deploying across ap-south-1a and ap-south-1b eliminates single AZ as a single point of failure.

Final Takeaway

This architecture demonstrates:

High Availability

Multi-AZ deployment across two AWS availability zones

Fault Tolerance

Automatic detection and replacement of unhealthy instances

Auto Scaling

Dynamic capacity management based on demand and health

Automatic Recovery

Zero-touch self-healing via ASG and ALB health checks

"This project reflects a production-style AWS architecture used in real-world systems."
Instances are treated as disposable, ensuring resilience and automated recovery.
Replicates patterns used in enterprise-grade cloud deployments for reliability and zero-downtime operations.